

Exercise JA

Q1. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is -

- (A) 22 (B) 23 (C) 24 (D) 25

Ans. D

Sol. $a_1, a_2, a_3, \dots$ be in H.P

$$\Rightarrow \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots \text{ be in A.P.}$$

$$\text{in A.P. } T_1 = \frac{1}{a_1} = \frac{1}{5} \text{ and } T_{20} = \frac{1}{a_{20}} = \frac{1}{25}$$

$$\frac{1}{25} = \frac{1}{5} + 19d \Rightarrow d = -\frac{4}{19 \times 25}$$

$$T_n = T_1 + (n-1)d < 0$$

$$\Rightarrow \frac{1}{5} - \frac{(n-1) \cdot 4}{19 \times 25} < 0 \quad \frac{1}{5} < \frac{4(n-1)}{25 \times 19}$$

$$\Rightarrow \frac{19 \times 25}{4} + 1 < n \Rightarrow \frac{19}{4} < n$$

$\Rightarrow$ least positive integer n is 25

Q2. A pack contains n cards numbered from 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k , then $k - 20 =$

Ans. 5

Sol. When 1 and 2 are removed from numbers 1 to n then we get maximum possible sum of remaining numbers and when $n-1, n$ are removed then we get minimum possible sum of remaining numbers.

$$\Rightarrow \frac{n(n+1)}{2} - (2n-1) \leq 1224 \leq \frac{n(n+1)}{2} - 3$$

$$\Rightarrow \begin{cases} n^2 + n - 2454 \leq 0 \\ n^2 - 3n - 2446 \leq 0 \end{cases}$$

$$\Rightarrow \begin{cases} n \geq 50 \\ n \leq 50 \end{cases} \Rightarrow n = 50$$

$$\Rightarrow \frac{50(50+1)}{2} - x - x - 1 = 1224$$

$$\Rightarrow x = 25$$

$\Rightarrow 25^{\text{th}}$ and 26^{th} cards are removed from pack

$$\Rightarrow k = 25 \Rightarrow k - 20 = 5$$

Q3. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) :

(A) 1056 (B) 1088 (C) 1120 (D) 1332

Ans. A, D

Sol.
$$S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2 = -1^2 - 2^2 + 3^2 + 4^2 - 5^2 - 6^2 + 7^2 + 8^2 + \dots$$

$$= (3^2 - 1) + (4^2 - 2^2) + (7^2 - 5^2) + (8^2 - 6^2) + \dots = 2 \underbrace{[4 + 6 + 12 + 14 + 20 + 22 + \dots]}_{2n \text{ terms}}$$

$$= 2[(\underbrace{4 + 12 + 20 + \dots}_{n \text{ terms}}) + (\underbrace{6 + 14 + 22 + \dots}_{n \text{ terms}})]$$

$$2 \left[\frac{n}{2} (4 \times 2 + (n-1)8) + \frac{n}{2} (2 \times 6 + (n-1)8) \right] = 2[n(4 + 4n - 4) + n(6 + 4n - 4)]$$

$$= 2(4n^2 + (4n + 2)n) = 2(8n^2 + 2n) = 4n(4n + 1)$$

(A) 1056 = 32×33 $n = 8$

(B) 1088 = 32×34

(C) 1120 = 32×35

(D) 1332 = 36×37 $n = 9$

Q4. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b+2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is :

Ans. 4

Sol. Let $b = ar, c = ar^2$ $r \Rightarrow$ is integers. Also $= ar + 2 \Rightarrow a + ar^2 = 2ar + 6$

$$\Rightarrow a(r-1)^2 = 6 \Rightarrow r \text{ must be } 2 \text{ and } a = 6. \text{ Thus } \frac{a^2 + a - 14}{a + 1} = 4 \text{ Ans.}$$

Q5. Suppose that all the terms of an arithmetic progression (A.P.) are positive integers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is :

Ans. 9

Sol.
$$\frac{S_7}{S_{11}} = \frac{6}{11}$$

$$\frac{\frac{7}{2}[2a + 6d]}{\frac{11}{2}[2a + 10d]} = \frac{6}{11} \text{ Given } 130 < a + 6d < 140$$

$$\frac{7(a + 3)}{11(a + 5d)} = \frac{6}{11}$$

$$7a + 21d = 6a + 30d \Rightarrow 130 < 15d < 140$$

$$a = 9d$$

$$\text{Hence } d = 9$$

$$a = 81$$

$$\text{Hence } d = 9$$

Q6. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is

(A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{27}$ (D) $\frac{1}{25}$

Ans. C

Sol. $4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x} \geq 1$

By A.M. $\geq$ G.M.

$$\frac{4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x}}{3} \geq \left(4\alpha x^2 \cdot \frac{1}{2x} \cdot \frac{1}{2x}\right)^{1/3}$$

$$4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x} \geq 3(\alpha)^{1/3}$$

For the least value of α

$$3(\alpha)^{1/3} = 1 \Rightarrow \alpha = \frac{1}{27}$$

Q7. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$, then :

(A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$ (C) $s < t$ and $a_{101} = b_{101}$

(D) $s < t$ and $a_{101} < b_{101}$

Ans. B

Sol. If $\log_e b_1, \log_e b_2, \dots, \log_e b_{101} \rightarrow \text{A.P.}$

then $b_1, b_2, \dots, b_{101} \rightarrow \text{G.P.}$

Given $a_1 = b_1$ and $a_{51} = b_{51}$

$a_1, a_2, a_3, \dots, a_{50}, a_{51}$ are in Arithmetic progression.

Here, We can say $a_2, a_3, \dots, a_{50}$ are Arithmetic means between a_1 and a_{51} .

By A.M. $>$ G.M.

$$a_2 > b_2, a_3 > b_3, \dots$$

$$\Rightarrow s > t$$

Also a_1, a_{51}, a_{101} are in Arithmetic Progression and b_1, b_{51}, b_{101} are in Geometric Progression.

$$\Rightarrow b_{101} > a_{101} (\because a_1 = b_1, a_{51} = b_{51})$$

Q8. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, Then, the number of elements in the set $X \cup Y$ is _____.

Ans. 3748

Sol. $X = \{1, 6, 11, \dots\}$

$$Y = \{9, 16, 23, \dots\}$$

Common terms : 16, 51, 86,

$$t_p = 16 + (p-1)35 = 35p - 19 \leq 10086$$

$$\Rightarrow p \leq 288.7$$

$$\therefore n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$$

$$= 2018 + 2018 - 288$$

$$= 3748$$

Q9. Let α and β be the roots of $x^2 - x - 1 = 0$ with $\alpha > \beta$. For all positive integers n, define $a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, n \geq 1, b_1 = 1$ and $b_n = a_{n+1} + a_{n-1}, n \geq 2$ Then which of the following options is/are **CORRECT** :

(A) $b_n = \alpha^n + \beta^n$ for all $n \geq 1$ (B) $a_1 + a_2 + a_3 + \dots + a_n = a_{n+2} - 1$ for all $n \geq 1$

(C) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$ (D) $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$

Ans. C, A, B

Sol. $x^2 - x - 1 = 0$ roots are α and β $a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \alpha = \frac{1 + \sqrt{5}}{2}, \beta = \frac{1 - \sqrt{5}}{2}$ $a^2 - \alpha$

$$= \frac{\alpha^{n-1}(1 + \alpha^2) - \beta^{n-1}(1 + \beta^2)}{\alpha - \beta} = \frac{\alpha^{n-1}(\alpha+2) - \beta^{n-1}(\beta+2)}{\alpha - \beta} = \frac{\alpha^{n-1}\left(\frac{5+\sqrt{5}}{2}\right) - \beta^{n-1}\left(\frac{5-\sqrt{5}}{2}\right)}{\alpha - \beta} = \frac{\sqrt{5}\left(\alpha^{n-1}\left(\frac{5+\sqrt{5}}{2}\right) - \beta^{n-1}\left(\frac{5-\sqrt{5}}{2}\right)\right)}{\alpha - \beta}$$

$$= \sqrt{5}\left(\frac{\alpha^{n-1}(\alpha) + \beta^{n-1}(\beta)}{\alpha - \beta}\right) b_n = \frac{(\alpha^n + \beta^n)\sqrt{5}}{\sqrt{5}} = \alpha^n + \beta^n \text{ Option (2) } a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} = \frac{\alpha^{n-2}(\alpha^2) - \beta^{n-2}(\beta^2)}{\alpha - \beta}$$

$$= \frac{\alpha^{n-1} - \beta^{n-1}}{\alpha - 1} + \frac{\alpha^{n-2} - \beta^{n-2}}{\alpha - \beta} a_n = a_{n-1} + a_{n-2}$$

$$a_{n+2} = a_{n+1} + a_n \quad \left| \quad a_2 = \frac{\alpha^2 - \beta^2}{\alpha - \beta} = \alpha + \beta = 1 a_{n-1} = a_n + a_{n-1} a_n = a_{n-1} + a_{n-2} \right|$$

$$\text{Option(3)} \sum \frac{\alpha^n - \beta^n}{(\alpha - \beta)10^n} = \frac{1}{\alpha - \beta} \left(\sum \frac{\alpha^n}{10^n} - \sum \frac{\beta^n}{10^n} \right) = \frac{1}{\alpha - \beta} \left(\frac{\frac{\alpha}{10}}{1 - \frac{\alpha}{10}} - \frac{\frac{\beta}{10}}{1 - \frac{\beta}{10}} \right) = \frac{1}{\alpha - \beta}$$

$$= \frac{(10\alpha - 10\beta)}{(\alpha - \beta)(100 - (\alpha + \beta)10 + \alpha\beta)} = \frac{10}{100 - 10 + 1} = \frac{10}{89} \text{Option (4) } \sum \left(\frac{\alpha^n + \beta^n}{10^n} \right) = \sum \frac{\alpha^n}{10^n} + \sum$$

$$= \frac{\alpha(10 - \beta) + \beta(10 - \alpha)}{(10 - \alpha)(10 - \beta)} = \frac{10(\alpha - \beta) - 2\alpha\beta}{100 - (\alpha + \beta)10 + \alpha\beta} = \frac{10 + 2}{100 - 10 - 1} = \frac{12}{89}$$

- Q10. Let AP (a;d) denote the set of all the terms of an infinite arithmetic progression with first term a and common difference $d > 0$. If $AP(1; 3) \cap AP(2; 5) \cap AP(3; 7) = AP(a; d)$ then a + d equals

Ans. 157

Sol. $AP(1; 3) \cap AP(2; 5) \cap AP(3; 7) \Rightarrow AP(1, 4, 7, 10, \dots) \cap AP(2, 7, 12, 17, \dots) \cap AP(3, 10,$

- Q11. Let m be the minimum possible value of $\log_3 (3^{y_1} + 3^{y_2} + 3^{y_3})$ where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^2)$ is _____.

Ans. 8

Sol.
$$\frac{3^{y_1} + 3^{y_2} + 3^{y_3}}{3} \geq \left(3^{y_1} 3^{y_2} 3^{y_3} \right)^{\frac{1}{3}}$$

$$3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3 \left(3^{\frac{y_1 + y_2 + y_3}{3}} \right)$$

$$3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3^4$$

$$\log_3(3^{y_1} + 3^{y_2} + 3^{y_3}) \geq 4$$

$$\Rightarrow m = 4$$

$$\frac{x_1 + x_2 + x_3}{3} \geq \left(x_1 x_2 x_3 \right)^{\frac{1}{3}}$$

$$3 \geq \left(x_1 x_2 x_3 \right)^{\frac{1}{3}}$$

$$27 \geq x_1 x_2 x_3$$

$$\log_3(27) \geq \log_3(x_1 x_2 x_3)$$

$$\Rightarrow M = 3$$

$$\log_2(m^3) + \log_3(M^2) = \log_2(64) + \log_3(9)$$

$$= 6 + 2 = 8$$

- Q12. Let $a_1, a_2, a_3, \dots$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, \dots$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c, for which the equality $2(a_1 + a_2 + \dots + a_n) = b_1 + b_2 + \dots + b_n$ holds for some positive integer n, is _____

Ans. 1

Sol. 2 (sum of first n -terms of AP) = (sum of first n -terms of GP)

$$\Rightarrow 2\left(\frac{n}{2}[2c + (n-1)2]\right) = c \cdot \left(\frac{2^n - 1}{2 - 1}\right)$$

$$\Rightarrow 2nc + 2n^2 - 2n = c2^n - c$$

$$c = \frac{2n^2 - 2n}{2^n - 2n - 1} = \text{Positive Integer (since } a_i \text{ \& } b_i \text{ are sequences of positive integers)}$$

Now for $n = 1$; $c = 0$

$$n = 2 ; c = -4$$

$$n = 3 ; c = 12$$

$$n = 4 ; c = \frac{24}{7}$$

$$n = 5 ; c = \frac{40}{21}$$

$$n = 6 ; c = \frac{60}{51}$$

And for all $n \geq 7$; $c < 1$

So no integral value of c can be obtained further.

So only possible value of $c = 12$.

Q13. Let $l_1, l_2, \dots, l_{100}$ be consecutive terms of an arithmetic progression with common difference d_1 , and let $w_1, w_2, \dots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2 , where $d_1 d_2 = 10$. For each $i = 1, 2, \dots, 100$, let R_i be a rectangle with length l_i , width w_i and area A_i . If $A_{51} - A_{50} = 1000$ and the value of $A_{100} - A_{90} = k$, then $\frac{k + 1100}{10000} = ?$

Ans. 2

Sol. $A_{51} - A_{50} = 1000$

$$\ell_{51} w_{51} - \ell_{50} w_{50} = 1000$$

$$(\ell_1 + 50d_1)(w_1 + 50d_2) - (\ell_1 + 49d_1)(w_1 + 49d_2) = 1000$$

$$(50 - 49)\ell_1 d_2 + (50 - 49)w_1 d_1 + (50^2 - 49^2)d_1 d_2 = 1000$$

$$\ell_1 d_2 + w_1 d_1 + 99d_1 d_2 = 1000$$

$$\ell_1 d_2 + w_1 d_1 = 10$$

Now $A_{100} - A_{90}$

$$= \ell_{100} w_{100} - \ell_{90} w_{90}$$

$$= (\ell_1 + 99d_1)(w_1 + 99d_2) - (\ell_1 + 89d_1)(w_1 + 89d_2)$$

$$= (99 - 89)\ell_1 d_2 + (99 - 89)w_1 d_1 + (99^2 - 89^2)d_1 d_2$$

$$= 10\ell_1 d_2 + 10w_1 d_1 + (10)(188)d_1 d_2$$

$$= 10 \times 10 + 100 \times 188$$

$$= 100(189)$$

$$= 18900 = k$$

$$\frac{18900 + 1100}{10000} = 2$$

- Q14. Let $a_1, a_2, a_3, \dots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3, \dots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Then, which of the following is/are TRUE ?

(A) $T_{20} = 1604$ (B) $\sum_{k=1}^{20} T_k = 10510$ (C) $T_{30} = 3454$ (D) $\sum_{k=1}^{30} T_k = 35610$

Ans. B, C

Sol. $a_1 = 7, d = 8$

$$T_{n+1} - T_n = a_n \forall n \geq 1$$

$$S_n = T_1 + T_2 + T_3 + \dots + T_{n-1} + T_n$$

on subtraction

$$T_n = T_1 + a_1 + a_2 + \dots + a_{n-1}$$

$$T_n = 3 + (n-1)(4n-1)$$

$$T_n = 4n^2 - 5n + 4$$

$$\sum_{k=1}^n T_k = 4 \sum_{k=1}^n n^2 - 5 \sum_{k=1}^n n + 4n$$

$$T_{20} = 1504$$

$$T_{30} = 3454$$

$$\sum_{k=1}^{30} T_k = 35615$$

$$\sum_{k=1}^{20} T_k = 10510$$

- Q15. Let $\overbrace{75 \dots 57}^r$ denote the $(r+2)$ digit number where the first and the last digits are 7 and the remaining r digits are 5. Consider the sum $S = 77 + 757 + 7557 + \dots + \overbrace{75 \dots 57}^{98}$. If

$S = \frac{\overbrace{75 \dots 57}^{99} + m}{n}$, where m and n are natural numbers less than 3000, then the value of $m + n$ is _____.

Ans. 1219

Sol.

$$S = 77 + 757 + 7557 + \dots + \overbrace{75 \dots 57}^{98}$$

$$= 7(10 + 10^2 + \dots + 10^{99}) + 50 \left(1 + 11 + \dots + \overbrace{111 \dots 1}^{98} \right) + 7 \times 99$$

$$= 70 \left(\frac{10^{99} - 1}{9} \right) + \frac{50}{9} [(10 - 1) + (10^2 - 1) + \dots + (10^{98} - 1)] + 7 \times 99$$

$$= 70 \left(\frac{10^{99} - 1}{9} \right) + \frac{50}{9} \left[10 \left(\frac{10^{98} - 1}{9} \right) - 98 \right] + 7 \times 99$$

$$= \frac{7 \times 10^{100}}{9} - \frac{70}{9} + \frac{50}{9} \left[\frac{10^{99} - 1 - 9}{9} - 98 \right] + 7 \times 99$$

$$= \frac{7 \times 10^{100}}{9} - \frac{70}{9} + \frac{50}{9} \left[\overbrace{111 \dots 1}^{99} - 99 \right] + 7 \times 99$$

$$= \frac{7 \times 10^{100} - 70 + \overbrace{555 \dots 50}^{99}}{9} - 550 + 693$$

$$= \frac{\overbrace{7555 \dots 5}^{99} - 70 + 143 \times 9}{9}$$

$$= \frac{\overbrace{755 \dots 57}^{99} + 1210}{9}$$

$$m + n = 1219$$